Ensemble Methods

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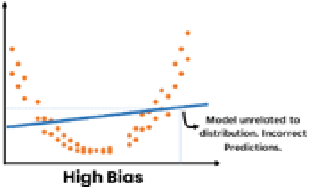
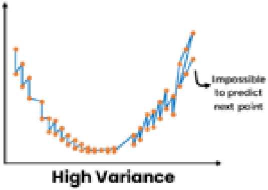
Outline

- Introduction to Ensembles
- Bagging and the Bootstrap
- Random Forests
- Boosting and its variants

Weak learners

- Models that perform only slightly better than random guessing are called *weak learners* (Bishop 2007).
- Weak learners low predictive accuracy may be due, to the predictor having high bias or high variance.





Trees may be weak learners

- Trees use to be sensitive to small changes in training data which lead to very different tree structure.
- This implies predictions are highly variable
- This may be explained because they are *greedy algorithms* making locally optimal decisions at each node without considering the global optimal solution.
    - This can lead to suboptimal splits and ultimately a weaker predictive performance.

There's room for improvement

- In many situations trees may be a good option (e.g. for simplicity and interpretability)
- But there are issues that, if solved, may improve performance.
- It is to be noted, too, that these problems are not unique to trees.
    - Other simple models such as linear regression may be considered as weakl learners in may situations.

The bias-variance trade-off

- When we try to improve weak learners we need to deal with the bias-variance trade-off.

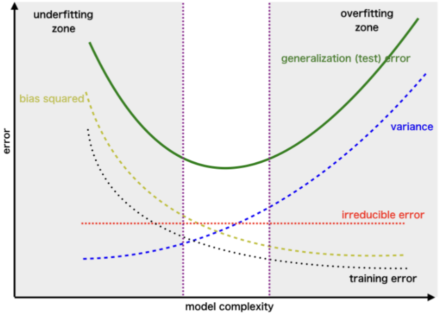


Figure 1: The bias-variance trade-off

How to deal with such trade-off

- How can a model be made less variable or less biased without this increasing its bias or variance?
- There are distinct appraches to deal with this problem
  - Regularization,
  - Feature engineering,
  - Model selection
  - Ensemble learning

Ensemble learners

- Ensemble learning takes the approach popularly known as "*the wisdom of crowds*".
- Predictors, also called, **ensembles** are built by fitting repeated (weak learners) models on the same data and combining them to form a single result.
- Ensemble learners tend to improve the results obtained with the weak learners they are made of.

Ensemble methods

- If we rely on how they deal with the bias-variance trade-off we can consider distinct groups of ensemble methods:
  - Bagging
  - Boosting
  - Hybrid learners

Ensemble methods cheatsheet

Aggregating Trees

One type of ensembles is made by constructing multiple trees using different types of subsets.

- **Bagging** (*B*ootstrap *agg*regat*ing*)
  - Random subsets *of observations*.
- **Random Forests** ($\simeq$ Bagging + Feature Selection)
  - Random subsets of observations
  - Random subsets of features at each split

These ensembles reduce variance compared to individual decision trees.

Boosting & Stacking

- **Boosting** or **Stacking** combine distinct predictors to yield a model with an increasingly smaller error, and so reduce the bias.
- They differ on if do the combination
  - sequentially (1) or
  - using a meta-model (2) .

Hybrid Techniques

- **Hybrid techniques** combine approaches in order to deal with both variance and bias.
- The approach should be clear from their name:
  - *Gradient Boosted Trees with Bagging*
  - *Stacked bagging*

Bagging: bootstrap aggregation

- Decision trees suffer from high variance when compared with other methods such as linear regression, especially when n/p is moderately large.
- Given that this is intrinsic to trees, Breiman (1996) sugested to build multiple trees derived from the same dataset and, somehow, average them.

Averaging decreases variance

- Bagging relies, informally, on the idea that:
  - given $X \sim F()$, s.t. $Var_F(X) = \sigma^2$,
  - given a s.r.s. $X_1, ..., X_n$ from F then
  - if $\overline{X} = \frac{1}{N} \sum_{i=1}^{n} X_i$ then $var_F(\overline{X}) = \sigma^2/n$.
- That is, *relying on the sample mean instead of on simple observations, decreases variance by a factor of* n.
- BTW this idea is still (approximately) valid for more general statistics where the CLT applies.

What means *averaging trees*?

Two questions arise here:

1. How to go from X to $X_1, ..., X_n$?
   - This will be done using *bootstrap resampling*.
2. What means "averaging" in this context.
   - Depending on the type of tree:
     - Average predictions for regression trees.
     - Majority voting for classification trees.

*BAGG*ING: Bootstrap Aggregation

- Breiman (1996) combined the ideas of:
  - Averaging provides decreased variance estimates,
  - Bootstrap provides multiple (re)samples.
- He suggested: **b**ootstrap **agg**regat**ing** :
  - Take resamples from the original training dataset
  - Learn the model on each bootstrapped training set
  - Get a prediction $\hat{f}^{*b}(x)$ from each model.
- Averaging predictions from single bootstrap models improves single prediction/classification.

Bagging prediction/classifier

- For regression (trees) the **bagged estimate** is the average prediction at x from these B trees.

$$\hat{f}_{bag}(x) = \frac{1}{B}\sum_{b=1}^{B}\hat{f}^{*b}(x)$$

- For classification (trees) the **bagged classifier** selects the class with the most "votes" from the B trees:

$$\hat{G}_{bag}(x) = \arg\max_{k} \hat{f}_{bag}(x).$$

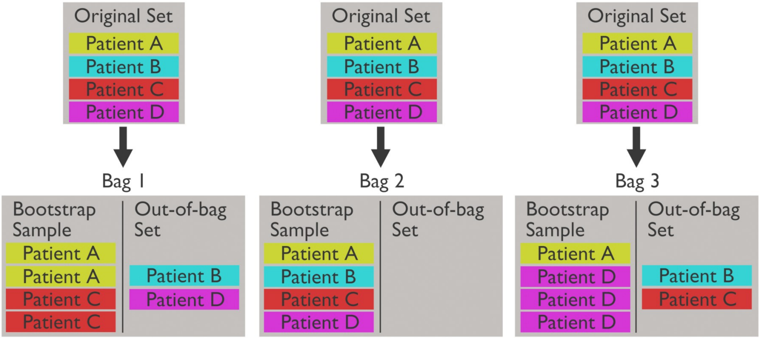
Resampling *based* estimators

- The bootstrap was introduced as a way to provide standard error estimators.
- When used to compute $\hat{f}_{bag}(x)$ or $\hat{G}_{bag}(x)$, as described above, it provides direct estimators of a characteristic, not of their standard errors.
- However, the bagging process *can also provide resampling based estimates of the precision of these estimators*.

Out-Of-Bag observations

- Every time a bootstrap sample is drawn *with replacement*, some observations are not selected.
- For a given tree, these omitted observations are called *out-of-bag* (OOB) observations.
- On average, each bootstrap sample leaves out about one third of the observations: $\left(1 - \frac{1}{n}\right)^n \approx e^{-1} \approx 0.368$

Out-Of-Bag samples



Out-Of-Bag error estimates (1)

- Since each out-of-bag set is not used to train the model, it can be used to evaluate performance.
- The key idea is that OOB status is defined *for each observation and each tree*.
  - Each tree is trained using its own bootstrap sample.
  - Therefore, each tree has its own OOB observations.
  - For a given observation i, we can identify the trees that were trained without using observation i.

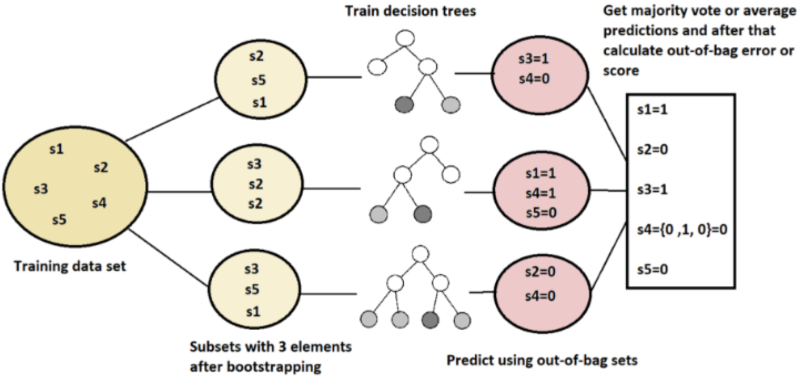
Out-Of-Bag error estimates (1)

OOB prediction for observation i is obtained as:

1. Find all trees for which observation i was out-of-bag.
2. Predict y_i using only those trees.
3. Combine these predictions by averaging (regression) or majority vote (classification).
4. Compare OOB prediction with true value y_i.
5. Aggregate errors over all observations.

OOB error is an **internal estimate** of the test error.

Illustration of OOB EE



Source:
https://www.baeldung.com/cs/random-forests-out-of-bag-error

Bagging in R

We use the `AmesHousing` dataset on house prices in Ames, IA, to predict the "Sales price" of houses.

1. Start by splitting the dataset in test/train subsets
2. Build a bagged tree on the train subset and evaluate it on the test subset.
3. Interpret the results using "Variable importance"

We use `randomForest` package setting `mtry` to total number of features.

Bagging - Prepare data

```
# Prepare "clean" dataset from raw data
ames <- AmesHousing::make_ames()
# Scale response variable to improve readability
ames$Sale_Price <- ames$Sale_Price/1000
# Split in test/training
set.seed(123)
train <- sample(1:nrow(ames), nrow(ames)/2)
# split <- rsample::initial_split(ames, prop = 0.7,
#                                 strata = "Sale_Price")
ames_train  <- ames[train,]
ames_test   <- ames[-train,]
```

Bagging - Build bag of trees

```r
library(randomForest)
set.seed(12543)
bag.Ames <- randomForest(Sale_Price ~ .,
                         data = ames_train,
                         mtry = ncol(ames_train)-1,
                         ntree = 100,
                         importance = TRUE)
```

Bagging - Results

```r
print(bag.Ames )
```

```
Call:
 randomForest(formula = Sale_Price ~ ., data = ames_train,
               Type of random forest: regression
                     Number of trees: 100
No. of variables tried at each split: 80

          Mean of squared residuals: 875.2949
                    % Var explained: 87.28
```

The reported MSE is the OOB estimate of prediction error

```
[1] 875.2949
```

External test error

Although OOB error provides an internal estimate of prediction error, we can also evaluate performance using an independent test set.

```
yhat.bag <- predict(bag.Ames, newdata = ames_test)
MSE= mean((yhat.bag -ames_test$Sale_Price)^2)
print(MSE)
```

- Here, predictions are made on completely unseen observations.
- This is a standard test error estimate, not an OOB estimate.

Interpetability: The "achiles heel"

- Trees may have a straightforward interpretation,
  - Plotting the tree provides information about
    - which variables are important
    - how they act on the prediction
- Ensembles are less intuitive because
  - there is no consensus tree.
  - not clear which variables are most important.

Feature importance from Trees

- To measure feature importance, the reduction in the loss function (e.g., SSE) attributed to each variable at each split is tabulated.
- The total reduction in the loss function across all splits by a variable are summed up and used as the total feature importance.

Feature importance for Ensembles

- Variable importance measures can be extended to an ensemble simply by adding up variable importance over all trees built.

	%IncMSE	IncNodePurity
Gr_Liv_Area	28.539804	1.104450e+12
Overall_Qual	28.512566	6.776106e+12
Neighborhood	16.369936	1.526868e+12
Total_Bsmt_SF	11.254286	6.535928e+11
First_Flr_SF	11.077080	4.336448e+11
MS_SubClass	9.303874	8.894513e+10
Garage_Area	8.390321	1.105019e+11
Overall_Cond	6.517383	4.146669e+10
Year_Remod_Add	6.440552	5.582591e+10
Kitchen_Qual	6.340537	4.133352e+10

Random forests: decorrelating predictions

- Bagged trees, based on re-samples (of the same sample) tend to be highly correlated.
- Leo Breimann, again, introduced a modification to bagging, he called *Random forests*, that aims at decorrelating trees as follows:
  - When growing a tree from one bootstrap sample,
  - At each split use only a randomly selected *subset of predictors*.

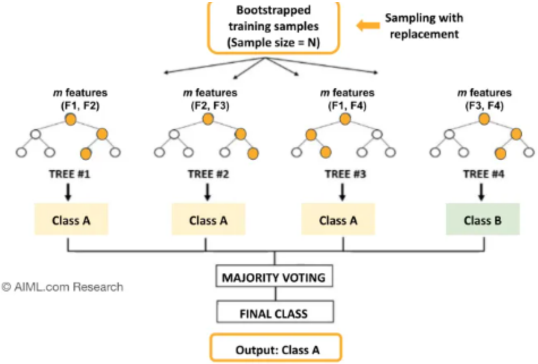
Split variable randomization

- While growing a decision tree, during the bagging process,
- Random forests perform *split-variable randomization*:
  - each time a split is to be performed,
  - the search for the split variable is limited to a random subset of m_{try} of the original p features.

Random forests

Training Data
(Sample size, N= 6, No. of features, F = 4)

F1	F2	F3	F4	Y
2.1	0	400	-9	A
3.0	1	890	-42	B
2.2	1	929	0	B
4.0	0	324	-23	A
3.5	1	333	-15	A
6.0	0	215	-9	A



Key parameters of Random Forest Model are: (a) Number of trees , (b) Maximum depth of the trees (c) Size of the random subset of features
In this example, No. of trees = 4, Depth = 2, and Feature subset size, m = 2 (no. of features/2)

Source: AIML.com research

How many variables per split?

- m can be chosen using cross-validation, but
- The usual recommendation for random selection of variables at each split has been studied by simulation:
  - For regression a typical default is $m = p/3$
  - For classification a typical default is $m = \sqrt{p}$.
- If $m = p$, we have bagging instead of RF.

Random forest algorithm (1)

```
1.  Given a training data set
2.  Select number of trees to build (n_trees)
3.  for i = 1 to n_trees do
4.  |  Generate a bootstrap sample of the original data
5.  |  Grow a regression/classification tree to the bootstrapped data
6.  |  for each split do
7.  |  | Select m_try variables at random from all p variables
8.  |  | Pick the best variable/split-point among the m_try
9.  |  | Split the node into two child nodes
10. |  end
11. | Use typical tree model stopping criteria to determine when a
    | tree is complete (but do not prune)
12. end
13. Output ensemble of trees
```

Figure 2: RF Algorithm, from ch. 11 in (Boehmke and Greenwell 2020)

Random forest algorithm (2)

Algorithm 17.1 RANDOM FOREST.

1 Given training data set $\boldsymbol{d} = (\boldsymbol{X}, \boldsymbol{y})$. Fix $m \leq p$ and the number of trees B.

2 For $b = 1, 2, \ldots, B$, do the following.

(a) Create a bootstrap version of the training data $\boldsymbol{d}_b^*$, by randomly sampling the n rows with replacement n times. The sample can be represented by the bootstrap frequency vector $\boldsymbol{w}_b^*$.

(b) Grow a maximal-depth tree $\hat{r}_b(x)$ using the data in $\boldsymbol{d}_b^*$, sampling m of the p features at random prior to making each split.

(c) Save the tree, as well as the bootstrap sampling frequencies for each of the training observations.

3 Compute the random-forest fit at any prediction point x_0 as the average

$$\hat{r}_{\mathrm{rf}}(x_0) = \frac{1}{B}\sum_{b=1}^{B} \hat{r}_b(x_0).$$

4 Compute the OOB_i error for each response observation y_i in the training data, by using the fit $\hat{r}_{\mathrm{rf}}^{(i)}$, obtained by averaging only those $\hat{r}_b(x_i)$ for which observation i was *not* in the bootstrap sample. The overall OOB error is the average of these OOB_i.

Out-of-the box performance

- Random forests tend to provide very good out-of-the-box performance, that is:
  - Although several hyperparameters can be tuned,
  - Default values tend to produce good results.
- Moreover, among the more popular machine learning algorithms, RFs have the least variability in their prediction accuracy when tuning (Probst, Wright, and Boulesteix 2019).

Out of the box performance

- A random forest trained with all hyperparameters set to their default values can yield an OOB RMSE that is better than many other classifiers, with or without tuning.
- This combined with good stability and ease-of-use has made it the option of choice for many problems.

Out of the box performance example

```
# number of features
n_features <- length(setdiff(names(ames_train), "Sale_Price

# train a default random forest model
ames_rf1 <- ranger(
  Sale_Price ~ .,
  data = ames_train,
  mtry = floor(n_features / 3),
  respect.unordered.factors = "order",
  seed = 123
)

# get OOB RMSE
(default_rmse <- sqrt(ames_rf1$prediction.error))
## [1] 24859.27
```

Tuning hyperparameters

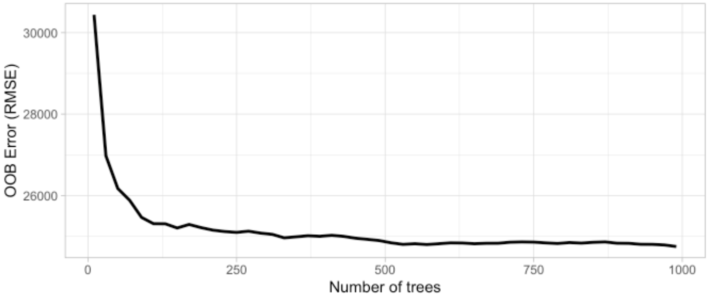
There are several parameters that, appropriately tuned, can improve RF performance.

1. Number of trees in the forest.
2. Number of features to consider at any given split (m_{try}).
3. Complexity of each tree.
4. Sampling scheme.
5. Splitting rule to use during tree construction.

1 & 2 usually have largest impact on predictive accuracy.

1. Number of trees

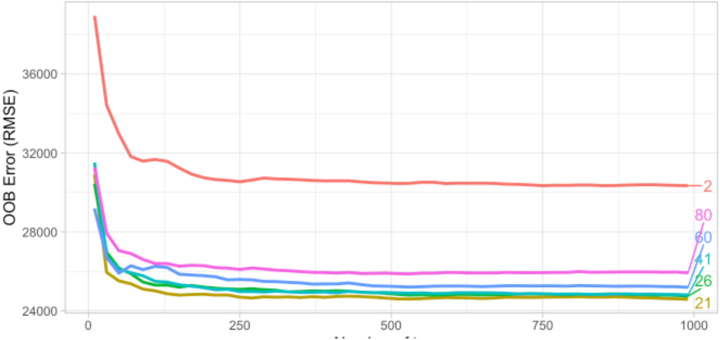
Start with $p \times 10$ trees and adjust as necessary



- The number of trees needs to be sufficiently large to stabilize the error rate.
- More trees provide more robust and stable error estimates
- But impact on computation time increases linearly with n_tree

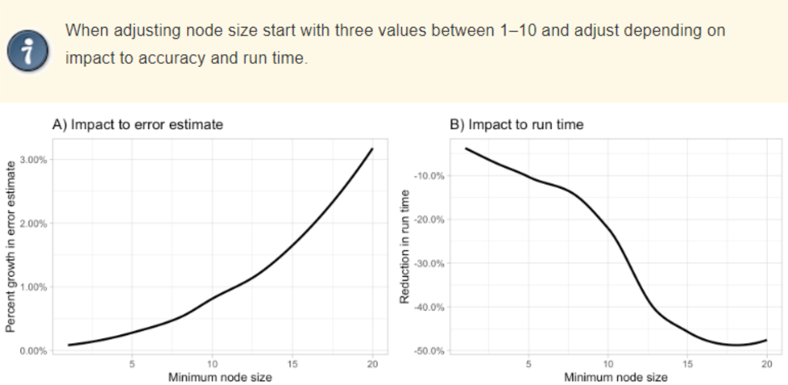
2. Number of features (m_{try}).

Start with five evenly spaced values of m_{try} across the range 2–p centered at the recommended default as illustrated in Figure 11.2.



- m_{try} helps to balance low tree correlation with reasonable predictive strength.
- Sensitive to total number of variables. If high /low, better

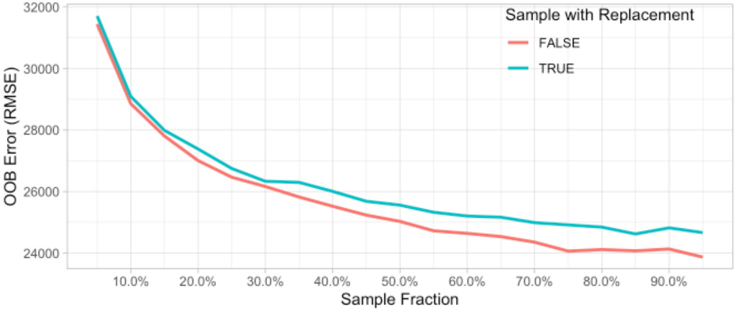
3. Complexity of each tree.



- The complexity of underlying trees influences RF performance.
- Node size has strong influence on error and time.

4. Sampling scheme.

Assess 3–4 values of sample sizes ranging from 25%–100% and if you have unbalanced categorical features try sampling without replacement.



- Default: Bootstrap sampling with replacement on 100% observations.
- Sampling fraction and replacement strategy influence diversity and prediction error.

5. Splitting rule

- *Default splitting rules* favor features with many splits, potentially biasing towards certain variable types.
- *Conditional inference trees* offer an alternative to reduce bias, but may not always improve predictive accuracy and have longer training times.
- *Randomized splitting rules*, like *extremely randomized trees* (which draw split points completely randomly), improve computational efficiency but may not enhance predictive accuracy and can even reduce it.

Tuning hyperparameters

- RF are a good example of a common situachion in ML:
  - As the number of parameter increases,
  - finding their optimal values requires more effort
  - and can even become computationally unfeasible.
- As more complex algorithms with greater number of hyperparameters are introduced tuning strategies should also be considered.

Tuning strategies

- **Grid Search**: Systematically searches through (*all possible combinations*) a predefined grid of hyperparameter values to find the combination that maximizes performance.
- **Random Search**: Randomly samples hyperparameter values from predefined distributions. Faster than Grid Search, but less prone to find optimum.
- **Model-Based Optimization** leverages probabilistic models, often Gaussian processes, to model the objective function and iteratively guide the search for optimal hyperparameters.

Random forests in bioinformatics

- Random forests have been thoroughly used in Bioinformatics (Boulesteix et al. 2012).
- Bioinformatics data are often high dimensional with
  - dozens or (less often) hundreds of samples/individuals
  - thousands (or hundreds of thousands) of variables.

Application of Random forests

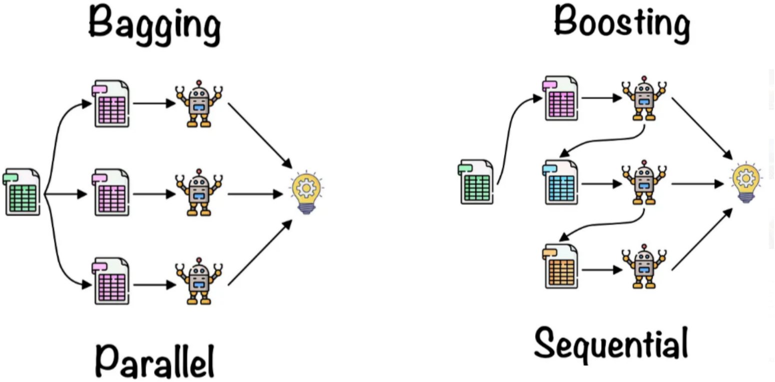
- Random forests provide robust classifiers for:
  - Distinguishing cancer from non cancer,
  - Predicting tumor type in cancer of unknown origin,
  - Selecting variables (SNPs) in Genome Wide Association Studies...
- Some variation of Random forests are used only for variable selection

Improving predictors iteratively

- The idea of *improving weak learners by aggregation* has moved historically along two distinct lines:

1. Build similar learners on resamples from the original sample and average the predictions.
   - This entails *Bagging* and *Random Forests*
2. Build a learner progressively, improving it at every step using weak learners, until the desired possible quality is obtained.
   - This is what *Boosting* is about.

Bagging vs Boosting



Source: Ensemble Learning: Bagging & Boosting

So bagging or boosting?

- Bagging/RF improves predictive performance by reducing **variance**.
  - *Averaging many unstable models* may produce a predictor with **smaller variance**.
- Averaging does not necessarily solve problems caused by **high bias**.
  - If individual models systematically underfit the data, averaging many of them may still produce an underfitted predictor.
  - Boosting aims to reduce **bias** by progressively improving the predictor.

A sequential correction

Boosting follows the following strategy:

- Models are built sequentially,
- Each new model attempts to correct errors made by previous ones.

Boosting constructs an additive model:

$$F(x) = f_1(x) + f_2(x) + f_3(x) + \dots$$

- $f_1(x)$ captures the main structure of the data,
- $f_2(x)$ corrects part of the remaining errors,
- $f_3(x)$ corrects the residual errors left
- and so on.

Final predictor is built incrementally through a sequence of weak learners.

General idea of Boosting

Boosting transforms many weak learners into a strong learner through *sequential error correction*.

1. It trains an initial weak learner.
2. Evaluates its errors.
3. Trains a new learner focused on correcting those errors.
4. Adds the new learner to the ensemble.
5. Repeats the process iteratively.

Boosting variants

- Different Boosting algorithms have been developed.
- They mainly differ in:
  - how the errors are defined and,
  - how next learner is constructed to reduce them.
- This leads to different Boosting methods such as:
  - AdaBoost,
  - Gradient Boosting,
  - XGBoost,
  - LightGBM.

AdaBoost

- It proceeds **Ada**ptatively by sequentially training weak,learners.
- Each new learner focuses increasingly on the difficult observations.
  - observations that are difficult to classify receive more attention,
  - while observations already classified correctly become less important.
- Final prediction combines all weak learners using a weighted vote.

Running Adaboost

At each iteration AdaBoost performs four main steps:

1. Fit a weak learner using the current observation weights.
2. Evaluate which observations are misclassified.
3. Increase the weights of difficult observations.
4. Assign a reliability weight to the learner itself.

As iterations proceed:

- difficult observations receive more attention,
- and accurate learners receive more influence in the final prediction.

The final classifier is therefore a weighted combination of weak learners.

Why do learner weights appear?

Initially, all observations receive the same weight, $w_i = \frac{1}{N}$

AdaBoost gives more influence to learners that achieve better classification performance.

If a learner has weighted error rate, ϵ_t, its contribution to the final ensemble is determined by:

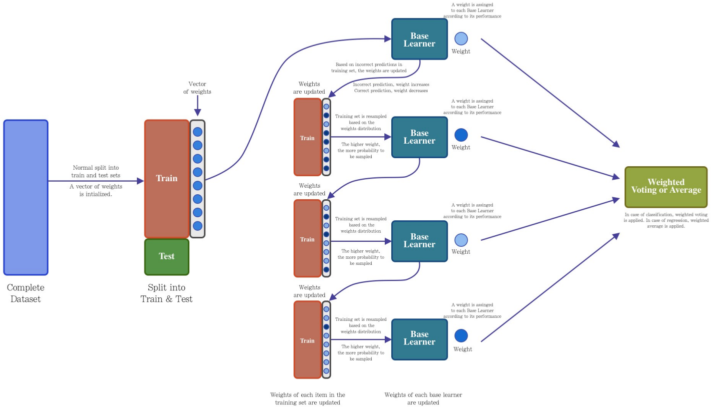
$$\alpha_t = \frac{1}{2} \log \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

Consequently:

- highly accurate learners receive larger weights,
- learners close to random guessing receive very small weights.

The final prediction is therefore dominated by the most reliable classifiers.

Adaboost Architecture

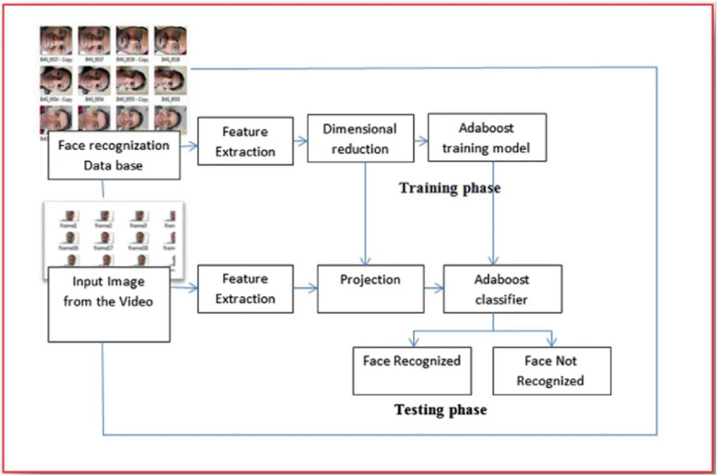


Source: Ensemble Learning: Bagging & Boosting

Adaboost algorithm

1. Initialize sample weights: $w_i = 1/N$, where $N = \#$ of training samples.
2. For each iteration $t = 1, 2, \dots, T$ do:
   a. Train weak classifier $h_t(x)$ on training set weighted by w_i.
   b. Compute the weighted error rate:
   $$\epsilon_t = \sum_{i=1}^{N} w_i I(y_i \neq h_t(x_i)) \Bigg/ \sum_{i=1}^{N} w_i$$
   c. Compute the classifier weight: $\alpha_t = \frac{1}{2} \log \frac{1-\epsilon_t}{\epsilon_t}$.
   d. Update the sample weights: $w_i \leftarrow w_i^{-\alpha_t I(y_i \neq h_t(x_i))}$.
   e. Normalize the sample weights: $w_i \leftarrow \frac{w_i}{\sum_{j=1}^{N} w_j}$.
3. Output the final classifier: $H(x) = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x)\right)$.

Adaboost applications



source: Evaluating the AdaBoost Algorithm for Biometric-Based Face Recognition

AdaBoost as loss minimization (1)

- AdaBoost is often presented as a heuristic algorithm,
- But it is also an optimization procedure: It can be shown to minimize the exponential loss:

$$L(y, F(x)) = e^{-yF(x)}$$

where:

- $y \in \{-1, +1\}$,
- and $F(x)$ is the additive ensemble model.

Adaboost as loss minimization (2)

- The interpretation described above is important because it shows that *AdaBoost is not simply an arbitrary reweighting strategy*.
- Instead, AdaBoost can be understood as
  - a sequential optimization procedure that
  - incrementally builds an additive predictive model
  - by reducing a **specific** loss function.
- This connects AdaBoost with the more general framework of Gradient Boosting where more general loss functions can be minimized.

Moving beyond AdaBoost

AdaBoost introduced the key idea of sequentially improving weak learners.

But it has important limitations:

- Originally designed mainly for classification,
- It relies on a specific exponential loss,
- It can become sensitive to noisy observations and outliers.

This motivated the development of a more general framework:

- instead of designing specific reweighting rules,
- directly minimize a loss function through iterative optimization.

This idea leads to Gradient Boosting.

Gradient Boosting

- Developed to overcome the limitations of Adaboost.
- Takes a different approach that can be linked with Optimization by Gradient Descent.
- Several advantages over Adaboost
  - Can handle continuous variables much better,
  - It is more robust to noisy data and outliers.
  - Can handle complex datasets and
  - Can continue to improve its accuracy even after Adaboost's performance has "plateaued".

A function optimization

Gradient Boosting views learning as an optimization problem.

Instead of directly searching for a single complex predictor, the model is *built incrementally* as:

$$F_M(x) = F_{M-1}(x) + \gamma_M h_M(x)$$

where:

- $F_{M-1}(x)$ is the current ensemble,
- $h_M(x)$ is a new weak learner,
- and γ_M controls how much the new learner contributes.

At each iteration, the algorithm searches for the weak learner that most reduces the current loss function.

The final model is constructed through a sequence of small corrective updates.

Relation with Gradient Descent

In gradient descent optimization, parameters are updated in the direction that most decreases the loss function:

$$\theta_{new} = \theta_{old} - \eta \nabla L$$

Gradient Boosting *applies the idea to predictive functions*.

Instead of updating numerical parameters the algorithm *updates the prediction function itself*.

At each iteration, a new weak learner is fitted to approximate the direction of steepest loss reduction.

Residual fitting as a special case

- For squared error loss, $\{L(y, F(x)) = (y - F(x))^2\}$ the negative gradient becomes: $\{-\partial L/\partial F = y - F(x)\}$, which *corresponds exactly to the residuals*.
- Therefore, for least-squares regression, fitting the negative gradient, and fitting the residuals, are equivalent procedures.
- This explains why Gradient Boosting is often described as "iteratively fitting residuals".
- However, for other loss functions the negative gradient is no longer the residual, and Gradient Boosting provides a much more general framework.

Gradient Boosting algorithm

1. Initialize model with constant prediction, $F_0(x)$.
2. For each iteration $m = 1, \dots, M$:
   a. Compute the negative gradient of the loss function.
   b. Fit a weak learner $h_m(x)$ to the negative gradient.
   c. Find the optimal step size:γ_m.
   d. Update the ensemble

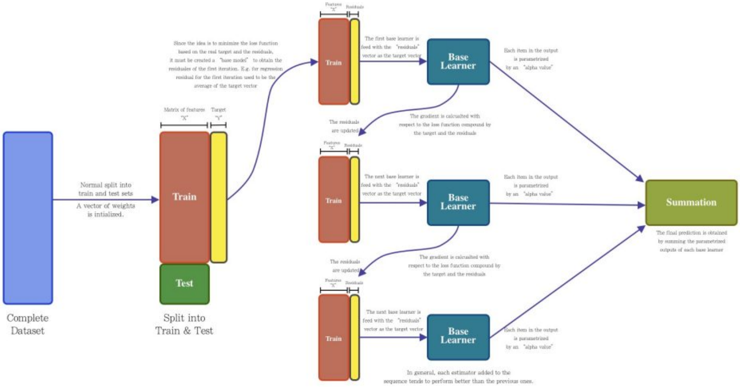
$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

3. Output the final ensemble model.

Gradient Boosting pseudo-code

1. Initialize the model with a constant value: $f_0(x) = \frac{1}{n}\sum_{i=1}^{n} y_i$
2. For $t = 1$ to T:
   a. Compute the negative gradient of the loss function at the current fit: $r_{ti} = -\frac{\partial L(y_i, f_{t-1}(x_i))}{\partial f_{t-1}(x_i)}$
   b. Train a new model to predict the negative gradient values:
   $$h(x;\theta_t) = \arg\min_h \sum_{i=1}^{n} (r_{ti} - h(x_i;\theta))^2$$
   c. Compute the optimal step size:
   $$\gamma_t = \arg\min_\gamma \sum_{i=1}^{n} L(y_i, f_{t-1}(x_i) + \gamma h(x_i;\theta_t))$$
   d. Update the model: $f_t(x) = f_{t-1}(x) + \gamma_t h(x;\theta_t)$
3. Output the final model: $F(x) = f_T(x)$

Gradient boosting architechture



Source: Ensemble Learning: Bagging & Boosting

Gradient Boosting Variations

- **XGBoost**
  - Optimized implementation that uses regularization to control overfitting and provide better accuracy.
- **LightGBM**
  - Relies on a technique to reduce the number of samples used in each iteration.
  - Faster training, good for large datasets.
- **CatBoost**
  - specifically designed to handle categorical variables efficiently.

Boosting applications

- Fraud Detection
- Image and Speech Recognition
- Anomaly Detection
- Medical Diagnosis
- Amazon's recommendation engine
- Models that predict protein structures from amino acid sequences
- Pattern identification in fMRI brain scans.

Advantages of Boosting

- Boosting, like other Ensemble methods, improves the accuracy of weak learners and achieve better predictive performance than individual models.
- Boosting also reduces overfitting by improving the generalization ability of models.
- Available in many flavors,
- Strong experience in Real world applications and industry.

Limitations of Boosting

- Can be computationally expensive, especially when dealing with large datasets and complex models.
- Can be sensitive to noisy data and outliers,
- May not work well with certain types of data distributions.
- Not so good as "out-of-the-box": Requires careful tuning of hyperparameters to achieve optimal performance, which can be time-consuming and challenging.

Boosting application with R and Python

- Many R packages implement the many variations of boosting:
  - ada,
  - adabag,
  - mboost,
  - gbm,
  - xgboost
  - An interesting option is to rely on the caret package which allows to run the distinct methods with a common interface.
- In Python we usually rely on scikit-learn libraries although there are many alternative implementations
  - Scikit-learn ensemble
  - https://python-course.eu/machine-learning/boosting-algorithm-in-python.php

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